

DAY 78-97 / MASTER INDEX

Foundation Collection Notes Index

A guided map of 20 Markdown-sourced notes covering visualization, exploratory analysis, data preparation, databases, statistics, probability, and optimization foundations.

20

DAILY NOTES

236

NOTE PDF PAGES

3

LEARNING GROUPS

RECOMMENDED PATH

1**Visualize and inspect data**

Days 78-84: Matplotlib, Seaborn, EDA, train/test preparation, databases, SQL, and revision.

2**Work with data systems and numerical structures**

Days 85-91: SQL CRUD and JOINS, MySQL, MongoDB, Redis, algebra, vectors, matrices, and revision.

3**Build statistical and optimization foundations**

Days 92-97: descriptive statistics, probability, distributions, the Central Limit Theorem, derivatives, and gradient descent.

REVISION CHECKPOINTS

Use Day 84 to revise Days 78-83 and Day 91 to revise Days 85-90.

WEEK 12 / DAYS 78-84

Visualization, EDA, and data preparation

Matplotlib and Seaborn lead into exploratory data analysis, train/test preparation, database fundamentals, SQL filtering, and weekly revision.

DAY 78	Matplotlib Fundamentals FOCUS Matplotlib; figures; axes; line, bar, and scatter plots PDF Day-78-Matplotlib-Fundamentals.pdf	9 PAGES
DAY 79	Seaborn Fundamentals FOCUS Seaborn; DataFrame-based plotting; bar, scatter, and distribution plots PDF Day-79-Seaborn-Fundamentals.pdf	12 PAGES
DAY 80	Beginner Exploratory Data Analysis (EDA) FOCUS shape; columns; data types; missing values; summaries; distributions; relationships PDF Day-80-Exploratory-Data-Analysis.pdf	11 PAGES
DAY 81	Basic Train/Test Data Preparation FOCUS feature columns; label/target; X; y; training data; test data PDF Day-81-Train-and-Test-Data-Preparation.pdf	12 PAGES
DAY 82	Database Fundamentals FOCUS databases; tables; rows; columns; primary keys; relational databases PDF Day-82-Database-Fundamentals.pdf	11 PAGES
DAY 83	SQL Fundamentals — SELECT and WHERE FOCUS SELECT; FROM; WHERE; comparison operators; ORDER BY PDF Day-83-SQL-SELECT-and-WHERE.pdf	12 PAGES
DAY 84	Weekly Revision — Visualization, EDA and Data Preparation FOCUS revision of visualization, EDA, data preparation, databases, and SQL PDF Day-84-Weekly-Revision.pdf	13 PAGES

WEEK 13 / DAYS 85-91

Databases and mathematical representations

SQL CRUD and JOINS connect to MySQL, MongoDB, and Redis before algebra, vectors, matrices, and weekly revision.

DAY 85	SQL CRUD and Basic JOIN FOCUS INSERT; UPDATE; DELETE; INNER JOIN; foreign-key idea PDF Day-85-SQL-CRUD-and-JOIN-Basics.pdf	11 PAGES
DAY 86	MySQL Basics and Connecting Python to MySQL FOCUS MySQL; database server; connection; cursor; query; result PDF Day-86-MySQL-and-Python.pdf	14 PAGES
DAY 87	MongoDB Basics FOCUS NoSQL; databases; collections; documents; <code>_id</code> ; insert; find PDF Day-87-MongoDB-Basics.pdf	13 PAGES
DAY 88	Redis Basics FOCUS Redis; keys; values; SET; GET; cache and expiration concepts PDF Day-88-Redis-Basics.pdf	14 PAGES
DAY 89	Basic Algebra for Machine Learning FOCUS variables; constants; equations; coefficients; functions; linear equations PDF Day-89-Basic-Algebra-for-Machine-Learning.pdf	11 PAGES
DAY 90	Vectors and Matrices for Machine Learning FOCUS scalars; vectors; matrices; rows; columns; shape PDF Day-90-Vectors-and-Matrices.pdf	12 PAGES
DAY 91	Weekly Revision — Databases and Math Introduction FOCUS revision of SQL, MySQL, MongoDB, Redis, algebra, vectors, and matrices PDF Day-91-Weekly-Revision.pdf	11 PAGES

WEEK 14 / DAYS 92-97

Statistics, probability, and optimization foundations

Descriptive statistics and probability build toward distributions, the Central Limit Theorem, derivatives, gradients, and gradient descent.

DAY 92	Descriptive Statistics FOCUS mean; median; spread; variance; standard deviation PDF Day-92-Mean-Median-Variance-and-Standard-Deviation.pdf	10 PAGES
DAY 93	Probability Fundamentals FOCUS experiments; outcomes; events; coin, dice, and business examples PDF Day-93-Probability-Fundamentals.pdf	11 PAGES
DAY 94	Basic Probability Distributions FOCUS random variables; distributions; discrete and continuous values; normal distribution PDF Day-94-Probability-Distributions.pdf	11 PAGES
DAY 95	Central Limit Theorem — Beginner Introduction FOCUS population; samples; sample means; normal-distribution intuition PDF Day-95-Central-Limit-Theorem.pdf	10 PAGES
DAY 96	Derivatives for Machine Learning FOCUS functions; slope; positive, negative, and zero slopes; derivatives PDF Day-96-Derivatives.pdf	14 PAGES
DAY 97	Gradients and Gradient Descent — Beginner Introduction FOCUS gradients; downhill direction; learning rate; gradient descent updates PDF Day-97-Gradients-and-Gradient-Descent.pdf	14 PAGES

QUICK LOCATOR

Find the right note fast

Use this map when you know the concept you want to review but not the day that covers it.

WANT TO REVIEW...	CORE CONCEPTS	GO TO
Plotting and visualization	Matplotlib, Seaborn, figures, axes, and common plots	Days 78-79
Exploratory analysis and ML data	EDA, missing values, summaries, features, labels, train/test data	Days 80-81
Relational databases and SQL	tables, primary keys, SELECT, WHERE, CRUD, JOINS, MySQL connections	Days 82-86
NoSQL and caching	MongoDB documents and collections; Redis keys, values, cache, expiration	Days 87-88
Algebra and linear algebra	variables, equations, linear functions, scalars, vectors, matrices	Days 89-90
Descriptive statistics	mean, median, variance, standard deviation	Day 92
Probability and distributions	events, outcomes, distributions, normal distribution, sample means	Days 93-95
Derivatives and optimization	slope, derivatives, gradients, learning rate, gradient descent	Days 96-97
Weekly consolidation	visualization/data revision and databases/math revision	Days 84, 91

Collection span

Days 78 through 97, with one PDF generated from each source Markdown file.

Generated collection

20 note PDFs containing 236 note pages, plus this index.